Python: warm-up exercises on branching

This notebook was generated in **solutions** mode.

Exercise: Check if Number is Even

Exercise Description

Write a piece of code that checks whether a number `num` is even. If so, print a message.

Your solution

```
num = 42
if num % 2 == 0:
    print("Condition evaluates to True.")
    print("Number", num, "is even")
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code**

before running the cell containing the Python Tutor link.

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Even or Odd Message

Exercise Description

Write a piece of code that checks whether a number `num` is even. Print a message about the outcome (odd or even).

Your solution

```
num = 42
if num % 2 == 0:
    print("Condition evaluates to True.")
    print("Number", num, "is even")
else:
    print("Condition evaluates to False.")
    print("Number", num, "is odd")
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Conditional assignment

Exercise Description

Write a piece of code that checks whether a number `num` is odd or even. Assign a proper value (e.g., 'odd' or 'even') to a variable `result`, depending on the outcome of the condition Print the outcome at the end.

Your solution

```
num = 42
if num % 2 == 0:
    result = "even"
else:
    result = "odd"
print("Number", num, "is", result)
```

Debug your solution

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```

Exercise: Multiple Divisibility Check

Exercise Description

Write a piece of code that checks whether a number `num` is divisible by 2, 3, 5 and 7. Print a message when `num` is divisible by one of this number.

Your solution

```
num = 42
if num % 2 == 0:
    print("Number ", num, "is divisible by 2.")
if num % 3 == 0:
    print("Number ", num, "is divisible by 3.")
if num % 5 == 0:
    print("Number ", num, "is divisible by 5.")
if num % 7 == 0:
    print("Number ", num, "is divisible by 7.")
```

Debug your solution

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```
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```

Exercise: Nested Conditional Logic

Exercise Description

Write a piece of code that checks whether a number `num` is odd or even. Print a message informing the user. If odd, check whether the number is divisible by 5. Print a message informing the user. If even, check whether the number is divisible by 3. Print a message informing the user.

Your solution

```
num = 15
if num % 2 == 0:
    print("Number", num, "is even.")
    if num % 3 == 0:
        print("Number", num, "is also divisible by 3.")
    else:
        print("Number", num, "is not divisible by 3.")
else:
    print("Number", num, "is odd.")
    if num % 5 == 0:
        print("Number", num, "is also divisible by 5.")
    else:
        print("Number", num, "is not divisible by 5.")
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Dog Age Calculator

Exercise Description

Write a piece of code that asks the user to insert the dog's age in human's years and calculates a dog's age in dog's years. For the first two years, a dog year is equal to 10.5 human years. After that, each dog year equals 4 human years.

Your solution

```
age = int(input("Enter the dog's age in human years: "))
print("Inserted age:", age)

if age <= 2:
    dog_years = age * 10.5
else:
    dog_years = 2 * 10.5 + (age - 2) * 4

print("The dog's age in dog's years is:", dog_years)
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```